

Description of Additional Supplementary Files

File Name: Supplementary Data 1

Description: List of sequencing data generated for the 'Fuji' genome assembly.

File Name: Supplementary Data 2

Description: Summary statistics of the phased assemblies of 'Fuji' and published domesticated apple genome assemblies.

File Name: Supplementary Data 3

Description: Hemizygous regions in the diploid genome of 'Fuji'.

File Name: Supplementary Data 4

Description: List of the apple varieties used for genome sequencing.

File Name: Supplementary Data 5

Description: Summary statistics of germline variants across the 'Fuji' genome

File Name: Supplementary Data 6

Description: Genome-wide annotation of germline variants.

File Name: Supplementary Data 7

Description: Summary statistics of somatic variants identified in the 74 Fuji clonal varieties.

File Name: Supplementary Data 8

Description: Genome-wide annotation of somatic variants.

File Name: Supplementary Data 9

Description: Summary statistics of somatic variants detected in each of the 74 'Fuji' clonal varieties.

File Name: Supplementary Data 10

Description: Grouping of 'Fuji' clonal varieties based on their phenotypes and phylogenetic relationships.

File Name: Supplementary Data 11

Description: Candidate somatic variants specific to spur-type varieties in the spur-type clade

File Name: Supplementary Data 12

Description: Candidate somatic variants specific to early-maturing varieties in the early-maturing clade.

File Name: Supplementary Data 13

Description: Validation of somatic SNPs between spur-type and standard-type varieties.

File Name: Supplementary Data 14

Description: List of 'Fuji' clonally propagated varieties used for spur rate, crown width, and trunk height measurement

File Name: Supplementary Data 15

Description: Primers used in this study.